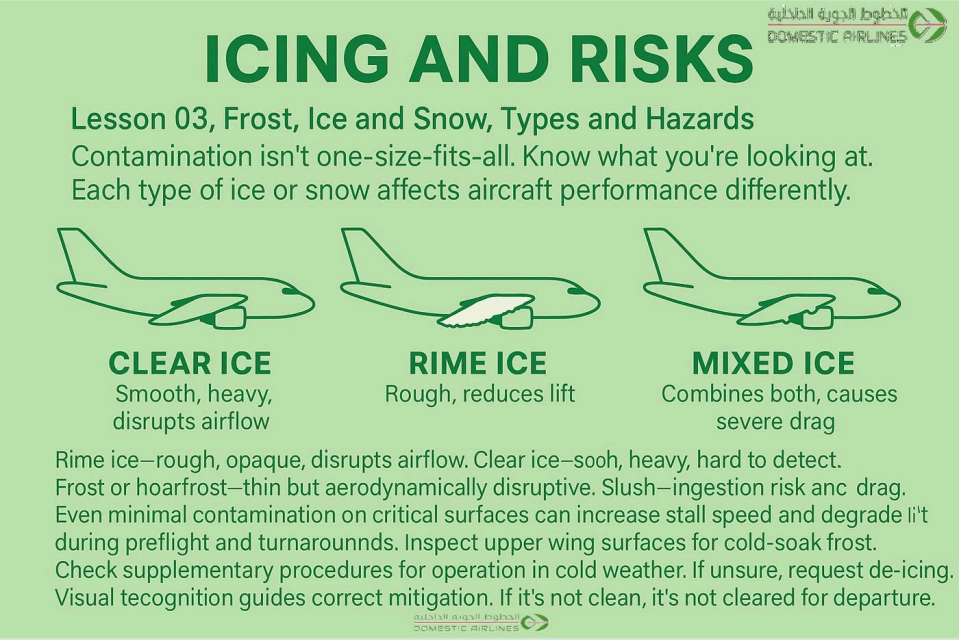
MODULE 1 · FOUNDATIONS

Lesson 03: Frost, Ice and Snow — Types and Hazards

Lesson 3 of 30 · about 8 min

LEARNING OBJECTIVES

- Identify the main forms of aircraft surface contamination.
- Describe the specific hazard of each contaminant.
- Recognise the conditions that produce clear ice on a cold-soaked wing.



Clear, rime and mixed ice — each affects the aircraft differently.

What counts as contamination

Any frost, ice, snow or slush that adheres to or sits on an aircraft surface is contamination. Even a thin, smooth-looking layer changes the shape and roughness of the surface. The main types are listed below.

Contaminant	Description	Specific hazard
Frost (hoar frost)	Thin, white crystalline deposit formed by direct deposition of water vapour.	Surface roughness. Even thin frost can significantly reduce lift and raise the stall speed.
Cold-soaked fuel frost (CSFF)	Frost or ice on wing surfaces above or below fuel tanks containing very cold fuel.	Can form at positive air temperatures and on a clear day; it often appears on the lower wing surface first.
Clear (glaze) ice	Transparent, smooth, hard ice from freezing rain or drizzle, or from rain on a cold-soaked wing.	Very difficult to see; may look like a wet wing. Can break off and be ingested by rear-mounted engines.
Rime ice	Opaque, white, rough ice from small supercooled droplets (freezing fog).	Builds on leading edges, probes, fan blades and propellers.

Dry snow	Fine snow that does not stick when the surface is cold. It can be blown off.	Sticks to a warm or wet surface and can melt and refreeze. Never assume it will blow off on take-off.
Wet snow	Snow with high liquid water content, near 0 °C.	Adheres readily, adds weight, and dilutes anti-icing fluid very quickly.
Slush	Water-saturated snow that splashes.	Thrown onto flaps, gear and engines during taxi; can refreeze in flight. On runways it causes drag and reduces braking.

Clear ice on a cold-soaked wing

After a long flight at altitude, the fuel in the wing tanks can remain far below 0 °C for hours. If the aircraft is then exposed to rain, drizzle, fog or high humidity on the ground, the water freezes on the cold wing skin and forms **clear ice**. This can happen at outside air temperatures of +5 °C or more.

WATCH FOR

The conditions are cold fuel (large fuel quantity left in the tanks after a long sector, or cold fuel uploaded), precipitation or high humidity, and air temperature near or above freezing. Clear ice cannot be reliably detected by eye — a **tactile check** by hand (or with an approved detection system) is required. See Lesson 12.

Acceptable deposits

Some AFMs permit specific, limited deposits. Typical examples are a thin layer of hoar frost on the upper fuselage, or CSFF on the lower wing surface in a defined area. These are **exceptions written into the AFM** for a particular type, with exact limits. If your AFM does not explicitly allow a deposit, it must be removed.

"Thin hoar frost" usually means a uniform white deposit thin enough that surface features — paint lines, markings and lettering — can still be seen through it.

KEY TAKEAWAYS

- All frost, ice, snow and slush on critical surfaces is contamination.
- Clear ice looks like a wet wing; it must be checked by touch.
- Cold-soaked fuel can produce frost or clear ice well above 0 °C.
- Only deposits explicitly permitted by the AFM, within its limits, are acceptable.

KNOWLEDGE CHECK

An aircraft arrives after a long sector with plenty of fuel remaining. It is +4 °C with light drizzle. What is the main concern?

A. Fuel freezing in the tanks

B. Clear ice forming on the cold-soaked wing

C. Brake overheating

D. No concern — the temperature is above freezing

Answer: B. Cold-soaked fuel keeps the wing skin below 0 °C; drizzle freezes on it as clear ice, which is hard to see and must be checked by touch.

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.